

[LAMP] Late Antiquity Modeling Project: Embodied Reconstructions of El Bagawat

1) **Project Title:** ML-Enhanced 3D Probabilistic Path-Tracing and Volumetric Viewshed Analysis

2) **Abstract / Project Summary:** Traditional archaeological modeling often relies on "least-cost paths" that only account for terrain steepness, treating human movement like water flow. Similarly, standard visibility analysis is typically 2-D and planimetric, ignoring building heights, window apertures, and translucent partitions. This project proposes an ML-enhanced framework for the Necropolis of El Bagawat. By using multispectral ground surface classification (**SAR-MS.tif**) to weight movement costs and incorporating building entrance orientations from architectural plans (**Site_Plan.pdf**), I will develop a probabilistic path ensemble that reflects social and physical realities. Furthermore, I will implement a true 3D ray-casting viewshed that accounts for 3D geometry and window openings, delivering high-fidelity GIS vector layers and 3D volumes to rewrite the embodied history of this late antique site.

3) **Contributor Name:** Romit Basak

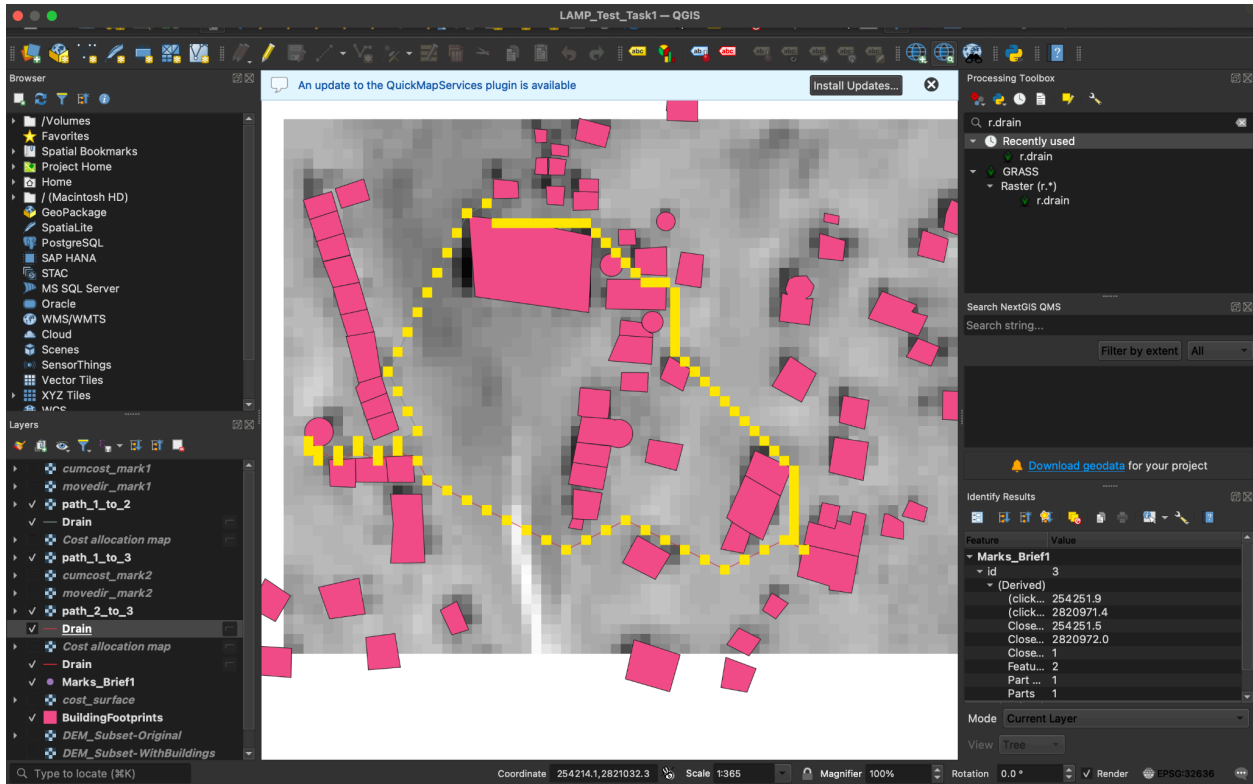
4) **Contributor Email and GitHub ID:** basak.r@northeastern.edu | <https://github.com/romit-basak>

5) **Potential Mentor(s):** Camille Leon Angelo (Alabama), Joshua Silver (KIT)

6) **Personal Background (Brief CV):** I am an MS Computer Science candidate at Northeastern University, working within the **Sustainability and Data Sciences Laboratory (SDS Lab)** under Prof. Dr. Auroop Ganguly. My interest in the Late Antiquity Modeling Project stems from a fascination with how computational vision can recover lost human experiences in historical landscapes. My current coursework in **CS5330 (Computer Vision)** focuses on 3D scene understanding, depth estimation, and shadow geometry—skills directly applicable to modeling the complex topography of El Bagawat. While my background is in ML and CV, I am committed to mastering **Space Syntax Analysis** and archaeological GIS workflows during the community bonding period to ensure my technical solutions serve the project's historical goals.

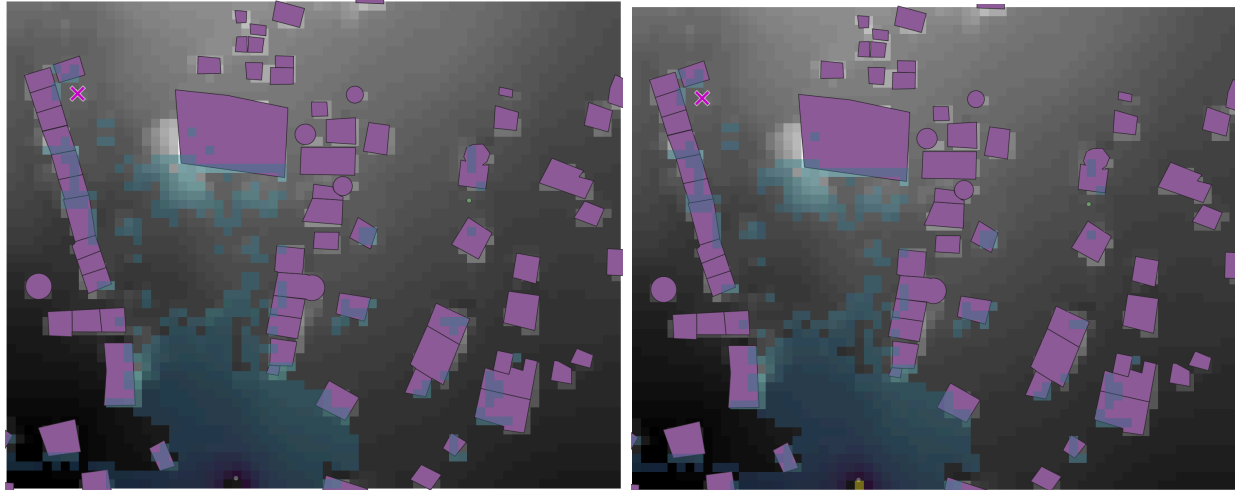
7) **Test Task Results:** I performed an initial baseline analysis using the provided El Bagawat dataset to identify the limitations of standard tools.

Task 1: Path Tracing Baseline * Methodology: I generated a cost surface by subtracting **DEM_Subset-Original.tif** from **DEM_Subset-WithBuildings.tif** to isolate architectural barriers. Using `r.cost` and `r.drain` in GRASS GIS, I traced paths between the termini defined in **Marks_Brief1.shp**. To ensure spatial accuracy, I verified entrance locations against **Site_Map_With_ROI.png**.



Critical Analysis: While the paths are geometrically "optimal," they are archaeologically naive. The current cost surface is uniform across open ground; it does not distinguish between surface materials—which can be inferred from the **SAR-MS.tif**—or the social "attraction" of building entrances marked in the **Site_Plan.pdf**. A person's path is dictated by the direction of an entrance and the texture of the ground, not just the slope. This motivates a multi-variable cost surface where SAR-MS.tif band ratios proxy surface hardness and the Site_Plan entrance vectors introduce directional asymmetry into the cost function.

Task 2: Visibility Baseline * Methodology: I executed `r.viewshed` from the locations provided in **Marks_Brief2.shp** using **DEM_Subset-WithBuildings.tif**. Crucially, I adjusted the observer height to **1.5m**, reflecting skeletal data for Late Antique Egyptian populations, rather than the 1.75m GIS default.



Teal pixels indicate areas visible from Mark 1 (1) left: at observer height 1.75m, (2) right: at observer height 1.5m

Critical Analysis: The 0.25m height difference noticeably restricted sightlines in some pixels. r.viewshed treats buildings as solid blocks. In reality, the window apertures and doorway orientations visible in **Site_Plan.pdf** would allow light and sight to permeate these structures—a feature the baseline tool currently misses.

8) Project Goals / Major Contributions: * **Multispectral Ground Classification:** Use the **SAR-MS.tif** bands to classify surface types (stone, sand, rubble). These will be used as differential cost weights, replacing the single-variable "steepness" model.

- **Entrance-Direction Weighted Model:** Extract building entrance orientations from **Site_Plan.pdf** and incorporate them as directional "cost attractors" to simulate how people naturally approach a doorway.
- **Probabilistic Path Ensemble:** Instead of a single deterministic line, I will use Monte Carlo sampling to generate a "probability heat map" of likely movement, exported as a GIS vector layer.
- **True 3D Ray-Casting Viewshed:** Implement a solution that accounts for building heights (**BuildingFootprints.shp**) and window openings. This will determine visibility from *inside* buildings looking out, and vice-versa, moving beyond 2D raster approximations.

9) Project Schedule:

9.1) Community Bonding Period (May 4 – May 24): * **May 4–10:** Study Space Syntax literature and depthmapX to align ML goals with archaeological theory.

- **May 11–17:** Analyze **SAR-MS.tif** band combinations for surface classification; extract entrance coordinates from **Site_Plan.pdf**.
- **May 18–24:** Align with mentors on success metrics (e.g., "archaeological plausibility" criteria).

9.2) Development Phase (May 25 – July 26): * **May 25–31: Deliverable:** Multispectral ground classification map derived from multispectral data.

- **June 1–7: Deliverable:** Weighted cost surface v2 (Surface Type + Slope + Entrance Vector).
- **June 8–14: Deliverable:** Probabilistic path ensemble via Monte Carlo sampling.
- **June 15–21: Deliverable:** GIS vector layer output for Task 1.
- **June 22–28: Deliverable:** 3D scene reconstruction (**DEM_Subset-WithBuildings.tif** + **BuildingFootprints.shp**).
- **June 29–July 5: Deliverable:** Ray-casting viewshed incorporating architectural apertures from site drawings.
- **July 6–12: Deliverable:** Comparison report: r.viewshed baseline vs. 3D Ray-casting.
- **July 13–19: Mid-term Evaluation:** MVP outputs for both tasks submitted as GIS vectors.
- **July 20–26: Deliverable:** Sensitivity analysis on observer height (1.5m vs 1.75m).

9.3) Project Completion (July 27 – August 16): * **July 27–Aug 2: Deliverable:** 3D viewshed volume rendering (Stretch Goal).

- **Aug 3–9: Deliverable:** Unified site analysis (Integrated Paths and Visibility).
- **Aug 10–16:** Final documentation, GitHub wiki, and performance report.

10) Planned GSoC Work Hours: * **Total:** 350 Hours (35 hours/week).

- **Timezone:** Based in Boston (EDT); work hours scheduled for 1:00 PM – 8:00 PM to overlap with mentors in Alabama (Morning) and Karlsruhe (Evening).
- **Commitments:** No planned absences or vacations.

11) Skill Set: * **Tools:** Python (PyTorch, Rasterio, GeoPandas, Shapely), GRASS GIS 7.8, QGIS.

- **Relevant Experience:** Developed the **Kolkata Urban Flood Mapping System** (<https://github.com/romit-basak/kolkata-flood-mapping>), demonstrating proficiency in multi-modal geospatial data and GNN/LSTM architectures.
- **CV Expertise:** 3D scene understanding and geometric reasoning from CS5330.
- **Self-Correction:** I acknowledge no prior experience with depthmapX but have already begun exploring the documentation to integrate its concepts into my Python-based workflow.